

```
(root@kali)-[~]
# ip addr show
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
  link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
  inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
  inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
  link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
  inet 172.20.10.2/28 brd 172.20.10.15 scope global dynamic noprefixroute eth0
    valid_lft 2023sec preferred_lft 2023sec
  inet6 fe80::e695:b332:6e37:8b33/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

```
(root@kali)-[~]
# sudo ip addr add 192.168.1.1/24 dev eth0

(root@kali)-[~]
#
```

```
(root@kali)-[~]
# nmcli radio wifi
enabled
```

```
(root@kali)-[~]
# sudo ip link set dev eth0 down

(root@kali)-[~]
# sudo ip link set dev eth0 address 00:11:22:33:44:55

(root@kali)-[~]
# sudo ip link set dev eth0 up
```

```
(root@kali)-[~]
# sudo dhclient -v eth0
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth0/00:11:22:33:44:55
Sending on   LPF/eth0/00:11:22:33:44:55
Sending on   Socket/fallback
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 8
DHCPOFFER of 172.20.10.3 from 172.20.10.1
DHCPREQUEST for 172.20.10.3 on eth0 to 255.255.255.255 port 67
DHCPACK of 172.20.10.3 from 172.20.10.1
bound to 172.20.10.3 -- renewal in 1594 seconds.
```

```
(root@kali)-[~]
# dig google.com NS

; <<>> DiG 9.20.15-2-Debian <<>> google.com NS
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 25194
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 4096
;; QUESTION SECTION:
;google.com.                IN      NS

;; ANSWER SECTION:
google.com.                4502    IN      NS      ns3.google.com.
google.com.                4502    IN      NS      ns4.google.com.
google.com.                4502    IN      NS      ns2.google.com.
google.com.                4502    IN      NS      ns1.google.com.

;; Query time: 124 msec
;; SERVER: 172.20.10.1#53(172.20.10.1) (UDP)
;; WHEN: Sat Feb 14 02:46:43 EST 2026
;; MSG SIZE rcvd: 111
```

```
(root@kali)-[~]
# dig google.com MX

; <<>> DiG 9.20.15-2-Debian <<>> google.com MX
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 63045
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 4096
;; QUESTION SECTION:
;google.com.                IN      MX

;; ANSWER SECTION:
google.com.                377     IN      MX      10 smtp.google.com.

;; Query time: 188 msec
;; SERVER: 172.20.10.1#53(172.20.10.1) (UDP)
;; WHEN: Sat Feb 14 02:48:20 EST 2026
;; MSG SIZE rcvd: 60
```

```
(root@kali)-[~]  
# l3afpad /etc/resolv.conf  
  
(root@kali)-[~]  
# cat /etc/resolv.conf  
# Generated by NetworkManager  
nameserver 172.20.10.1  
nameserver 8.8.8.8
```

EXERCISES

Before you move on to Chapter 4, try out the skills you learned from this chapter by completing the following exercises:

1. Find information on your active network interfaces.
2. Change the IP address on `eth0` to `192.168.1.1`.
3. Change your hardware address on `eth0`.
4. Check whether you have any available wireless interfaces active.
5. Reset your IP address to a DHCP-assigned address.
6. Find the nameserver and email server of your favorite website.
7. Add Google's DNS server to your `/etc/resolv.conf` file so your system refers to that server when it can't resolve a domain name query with your local DNS server.